# Outpost Zero

Pre-screening video script, Parts 1 to 6 (4:00)

NASA Space Apps 2026 | Build a Junior Astronaut Mission Trainer

| Part | Time | Topic | Length |
|---|---|---|---|
| **Part 1** | 0:00-0:40 | Introduction | 40 s |
| **Part 2** | 0:40-1:30 | Why it matters | 50 s |
| **Part 3** | 1:30-1:50 | Reveal | 20 s |
| **Part 4** | 1:50-2:40 | Features | 50 s |
| **Part 5** | 2:40-3:20 | Mission story (demo) | 40 s |
| **Part 6** | 3:20-4:00 | Impact, needs and credits | 40 s (ends by 3:58) |

**The 4:00 limit is a hard cutoff.** The six parts add up to exactly 240 seconds, so each part must finish on time. Part 6 ends at 3:58 to leave a 2-second buffer.

## Coordination notes across parts

- Part 1 and Part 2 both list a stars or Milky Way time-lapse. Use two different clips, or use it only once.
- Part 2 (real dust storm image), Part 3 (dim, dusty Mars surface footage) and Part 5 (2018 dust storm, Mars surface footage) all use NASA Mars material. Pick different images and clips for each part so nothing repeats.
- Part 1 asks “How do astronauts survive on another planet?” and Part 6 answers it in its last line, so the video opens and closes on the same question.
- Part 6 reuses the Part 2 split image (dream | NASA data) and the Part 5 lesson card, so keep those files.
- Parts 3 and 4 describe planned features (levels, crew, radio, badges, certificate). Part 5 shows the current prototype. Keep the wording as “adds” and “unlocks” for the planned features.
- Keep one running credits list for all six parts (the master list is on the last page).

# Part 1: Introduction (0:00-0:40)

about 38 seconds of narration. Goal: attention and authenticity. Win them with a story.

## 1. Images (one every 4-8 seconds)

- A black screen with the question “How do astronauts survive on another planet?” typing on, over the stars. Make the text yourself.
- A big white word card: “BRAVE.” on black. Make it yourself.
- A trade-off seesaw drawing: solar panel, battery, oxygen and water icons dropping onto a balance, which tilts each time. Make it yourself.
- A plain text wordmark “Outpost Zero” on a dark background with a soft orange glow. Make it yourself. Keep the full title card with the Mars horizon for Part 3, so the reveal stays there.
- A team card: your team name, then each member’s name. Make it yourself.

## 2. Video clips (4-5 seconds each, muted under your voice)

- Time-lapse of stars or the Milky Way as a calm opener (free stock). Use a different clip from Part 2’s stars time-lapse.
- NASA footage of an astronaut on a spacewalk or on the space station, 4-5 seconds (NASA media library).

## 3. Script (about 38 seconds)

| # | Image cue | Say |
|---|---|---|
| **1** | Stars clip, then astronaut clip, with the typed question | “When I was twelve, I looked up at the night sky and asked: how do astronauts survive on another planet?” |
| **2** | “BRAVE.” word card | “The answer I got was, ‘They’re brave.’” |
| **3** | Trade-off seesaw drawing | “But surviving on Mars is not about bravery alone. It is about decisions, made with limited supplies, where every choice helps one thing and hurts another.” |
| **4** | “Outpost Zero” wordmark | “That curious twelve-year-old is why we built Outpost Zero.” |
| **5** | Team card | “We are [Team name]: [Name 1], [Name 2], [Name 3], [Name 4], and [Name 5].” |

***Timing tip:*** *read it aloud with a timer. The script is about 80 words and should take 36 to 38 seconds, so Part 2 can start at 0:40. Under 34 seconds, slow down and pause for one second after line 2. Over 40 seconds, cut “made with limited supplies,” from line 3.*

## 4. Image and animation sequence

Add the assets in this order. Times are targets; adjust them to your voice recording, but keep the end at 0:39 or earlier.

| Time | Asset and format | Animation and transition | Source |
|---|---|---|---|
| **0:00-0:04** | Video clip, MP4, 1080p, 30 fps, muted.<br>Stars or Milky Way time-lapse. | Fade up from black over 1 s.<br>The question types on at speaking pace: "How do astronauts survive on another planet?"<br>Slow zoom 105% to 100%. | Pexels or Unsplash |
| **0:04-0:09** | Video clip, MP4, 4-5 s, muted.<br>NASA astronaut footage. | 0.5 s cross-dissolve from the stars.<br>The typed question stays on screen with a blinking cursor at the end. | NASA media library |
| **0:10-0:13** | Static PNG, 1920x1080.<br>"BRAVE." word card. | Hard cut from the clip.<br>Slow zoom in (100% to 110%).<br>At 0:13 the text fades to 30% opacity, to show the answer told them nothing. | You make it |
| **0:14-0:26** | Animated graphic, MP4 or GIF export, 1080p.<br>Trade-off seesaw. | Icons (solar, battery, oxygen, water) drop onto the seesaw one at a time and it tilts after each.<br>Text "helps one thing" and "hurts another" appears as you say those words. | You make it (Canva, PowerPoint or similar) |
| **0:27-0:31** | Static PNG, 1920x1080.<br>"Outpost Zero" wordmark. | 0.5 s dissolve from the seesaw.<br>Fade in with the letter spacing widening slightly over 2 s. | You make it |
| **0:32-0:39** | Static PNG, 1920x1080.<br>Team card. | Team name fades in first, then each name fades in (0.3 s each) as it is spoken.<br>Hold 0.5 s, then cut straight to Part 2 at 0:40. No fade longer than 0.3 s. | You make it |

***Format rules:*** *export at 1920x1080 (16:9). Use the same font and the same navy and orange palette as the other parts. Keep every transition under 0.5 s so the pace matches the other parts. Mute all clip audio under your voice.*

## Bonus: Pinterest searches (for ideas only)

**Photos:** kid looking at stars silhouette, typing text title card, balance scale illustration, team name card design, astronaut spacewalk.

**Videos:** Milky Way time-lapse, astronaut spacewalk, typewriter text animation.

## Sourcing note

The question card, word card, seesaw, wordmark and team card are your own work, so list the design tool you used. Take the stars clip from Pexels or Unsplash and the astronaut footage from NASA's own library after checking its usage rules. Add every source to the shared credits list.

## Note on wording

“When I was twelve” should be a true moment from the speaker’s life, because authenticity is what the first 45 seconds is judged on. Replace the wording with your real story if it differs. If you use a made-up child instead, say “imagine a twelve-year-old” and show a “Fictional story” tag on screen.

# Part 2: Why It Matters (0:40-1:30)

about 50-55 seconds

## 1. Images (one every 8-10 seconds)

- A child looking up at the night sky or a telescope. It shows the dream.
- A kid in a toy astronaut helmet, or a child's drawing of Mars.
- A real NASA photo of Mars, such as the Curiosity rover's view of Gale Crater.
- A real Mars dust storm image from orbit. It shows the planet fighting back.
- A classroom with a textbook, plus a confused student. It shows space STEM that is too simple or too hard.
- A simple drawing of one budget bar with power, air, water, food and radiation fighting for it. Make this one yourself.
- A split image: a child's dream on the left, a real NASA data chart on the right. This is the main message image.

## 2. Video clips (3-5 seconds each, muted under your voice)

- Kids watching a rocket launch.
- A kid playing a game or tablet with excited reactions.
- Mars rover driving or Mars surface footage from NASA.
- Time-lapse of stars or the Milky Way as a calm opener.

## 3. Script (with image cues, about 50-55 seconds)

| # | Image cue | Say |
|---|---|---|
| **1** | Child looking at the night sky | "Many kids dream of becoming astronauts. They look at the night sky and imagine walking on Mars." |
| **2** | Kid in a helmet | "These dreams matter. Kids are our future, and keeping their dreams alive helps them grow into builders." |
| **3** | Confused student | "But space-themed learning often oversimplifies real missions, or makes them too complex for a young learner to stay interested." |
| **4** | Budget drawing | "Real missions are about engineering trade-offs. Life support, radiation shielding, power and food production all share one limited budget. Spend more on one, and you have less for another." |
| **5** | Same drawing | "Few tools make these trade-offs both tangible and fun." |
| **6** | Mars outpost image | "We want students to run a Mars outpost and feel the decisions that make a mission fail or succeed." |
| **7** | Split image: dream and NASA data | "Outpost Zero connects a child's dream with real NASA data, like Mars radiation and ice maps." |

***Timing tip:*** *read it aloud with a timer. Under 48 seconds, slow down and pause after lines 2, 4 and 6. Over 55 seconds, cut the end of line 2.*

## Bonus: Pinterest searches

**Photos:** kid dreaming of space, child astronaut costume, kids looking at stars, child drawing Mars, Mars rover photo, Mars dust storm, classroom space lesson, students learning STEM.

**Videos:** kids watching rocket launch, child astronaut pretend play, stars time lapse, Mars surface video, kid playing tablet game, students excited science.

## Sourcing note

Pinterest is for ideas. Most images there belong to someone else, so use NASA's own media (check NASA's media usage rules) or free sources like Pexels and Unsplash for the final video. The challenge requires crediting all third-party assets and AI, so keep a list of every source as you collect it.

# Part 3: Reveal (1:30-1:50)

about 20 seconds

## 1. Images (one every 6-7 seconds)

- A title card, “OUTPOST ZERO”, over a Mars horizon and stars. Make this yourself.
- Four astronaut figures in front of a small habitat. Make it yourself or use AI, and credit the AI tool.
- A dashboard sketch with one bar turning orange and a red warning light. This is the intensity image. Make it yourself.
- A simple staircase of six steps, one situation or necessity per step. This shows the level-by-level structure. Make it yourself.

## 2. Video clips (3-4 seconds each, muted under your voice)

- A red warning light flashing (free stock clip).
- NASA Mars surface footage, dim and dusty.

## 3. Script (about 18-20 seconds)

| # | Image cue | Say |
|---|---|---|
| **1** | Title card | “Meet Outpost Zero. You command a four-person Mars outpost on a fixed budget.” |
| **2** | Crew + warning light | “The lights flicker. The oxygen drops. Four people are counting on you.” |
| **3** | Six-step staircase | “And you learn one situation at a time, level by level, so nothing clashes and everything sticks.” |

***Timing tip:*** *the pause after line 2 is what makes it feel tense, so keep it for about a second. Over 20 seconds, cut “on a fixed budget” from line 1, since Part 4 covers the budget.*

# Part 4: Features (1:50-2:40)

about 50 seconds

## 1. Images (one every 7-8 seconds)

- Four site cards: Gale Crater, Arcadia Planitia, Deuteronilus Mensae and an equatorial plain. Use a real NASA photo of Gale Crater for the first card and sketches for the rest.
- A budget screen sketch with solar panel, battery, greenhouse, shielding and spare part icons, and one budget bar. Make it yourself.
- Six level icons in a row: Power, Storms, Air, Radiation, Food and Water, Commander. Make it yourself.
- Four crew portraits, one with a status line like “Amira is cold”, plus a Mission Control radio bubble. Make it yourself.
- Three surprise-event icons: a meteor shower, a rover finding ice, a lucky clear day. Make it yourself.
- The mission patch (name, color, emblem) next to a badge like “Dust Fighter”. Make it yourself.
- The Junior Astronaut certificate with the mission patch on it. Make it yourself.

## 2. Video clips (3-5 seconds each, muted under your voice)

- A kid playing on a tablet with an excited reaction.
- A Mars rover driving (NASA).
- A meteor or shooting-star time-lapse.
- A child holding up a certificate or celebrating.

## 3. Script (about 50 seconds)

| # | Image cue | Say |
|---|---|---|
| **1** | Four site cards | “First, pick your landing site. Each one has different sunlight, ice and radiation.” |
| **2** | Budget screen | “Then spend a fixed budget on solar panels, batteries, a greenhouse, shielding and spare parts.” |
| **3** | Six level icons | “Each level adds one thing you must handle: Power, Storms, Air, Radiation, Food and Water. Then Commander, where it all comes together.” |
| **4** | Crew + radio bubble | “Meet your crew, hear Mission Control on the radio, and earn badges as you go.” |
| **5** | Surprise events | “And it’s not just six levels. Emergencies, meteor showers and surprise discoveries make every mission different.” |
| **6** | Six icons again | “Power, storms, air, radiation, food and water. Learn to handle every situation and necessity a crew needs, and you’re one step closer to being an astronaut.” |

| # | Image cue | Say |
|---|---|---|
| **7** | Patch + certificate | “Finish, and you earn your Junior Astronaut certificate. Keep learning, and keep dreaming.” |

***Timing tip:*** *read it aloud with a timer. Under 46 seconds, slow down and pause after lines 3 and 5. Over 54 seconds, cut line 4 down to “Meet your crew and earn badges as you go.”*

## Bonus: Pinterest searches (for ideas only)

**Photos:** Mars habitat concept art, astronaut crew illustration, game level progression UI, space mission patch design, junior astronaut certificate, meteor shower night sky.

**Videos:** kid playing tablet game, Mars rover driving, shooting star time-lapse, child holding certificate.

## Sourcing note

Most of the visuals above are meant to be drawn by you, so list the tool you used for each one. For anything AI-generated (crew art, for example), note the tool and what it made. The Gale Crater photo and rover footage should come from NASA’s own media (check its usage rules). Free stock like Pexels or Unsplash covers the rest. Keep one running credits list so the final slide is quick to build.

## Note on wording

Crew names, radio messages, badges, surprise events and the certificate are planned features, not in the current prototype. Since the video is concept-only that is fine, but say “adds”, “unlocks” and “earn” as the design, not as finished.

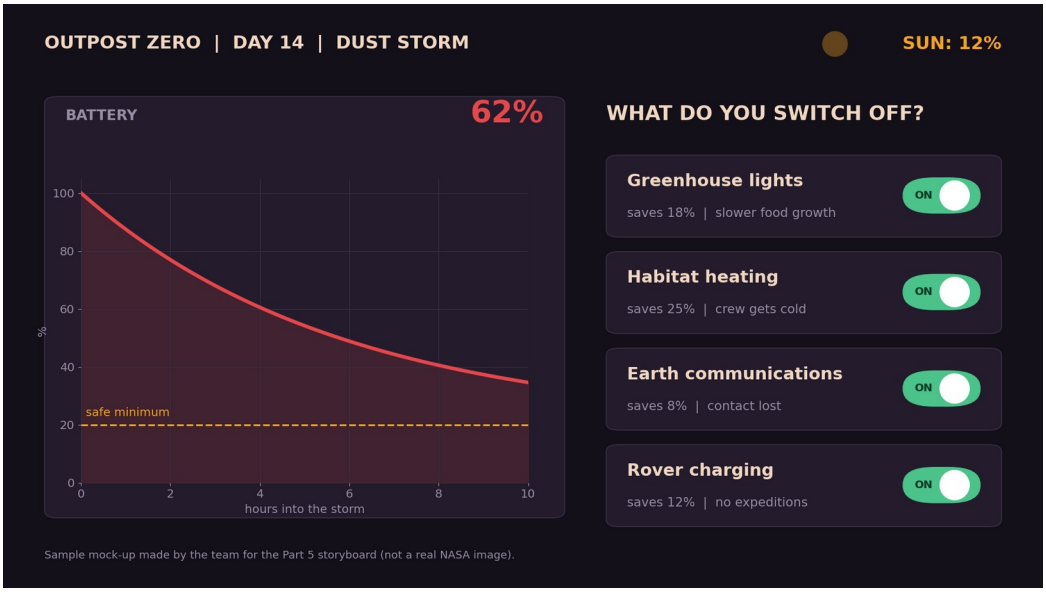
# Part 5: Mission Story (2:40-3:20)

Goal of this part (40 seconds): a dust storm hides the sun, the batteries drop, and the player must decide what to switch off. Then the game shows the "what went wrong" lesson.

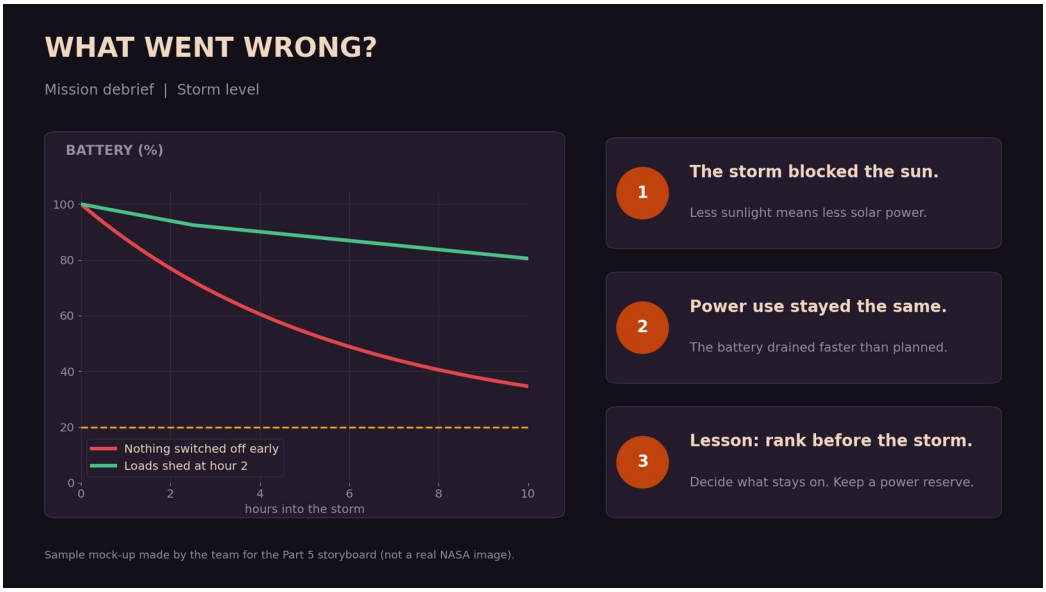
## 1. Images (one every 8-10 seconds, about 5 images)

| # | Image idea | What it shows | Where to get it |
|---|---|---|---|
| **1** | Mars before and after the 2018 dust storm (MRO camera, May vs July) | The storm swallowing a whole planet. Opens the part with a real NASA view. | NASA/JPL PIA22487<br>Credit: NASA/JPL-Caltech/MSSS |
| **2** | Opportunity rover's last images, with only a faint dot of Sun | "A dust storm hides the sun." Real proof that sunlight can almost disappear. | NASA PIA22930 and PIA22929<br>Credit: NASA/JPL-Caltech/Cornell/ASU |
| **3** | A solar panel or rover deck covered in dust | Dust on panels means less power. Links the storm to the battery problem. | Search NASA for "HiRISE Studies the Dust on InSight" or a rover selfie with a dusty deck. Check NASA's media usage rules. |
| **4** | Game decision screen: battery dropping, four switches (make this one yourself) | The moment of choice: "What do you switch off?" | Made by the team. Sample 1 below. |
| **5** | "What went wrong?" lesson card (make this one yourself) | Two battery lines (nothing switched off vs loads shed early) and three short lesson points. | Made by the team. Sample 2 below. |

Optional 6th image if you need more: a dim habitat interior in red emergency light (free stock from Pexels or Unsplash), for mood.




Sample 1 (image 4): decision screen. Numbers are placeholders; swap in the real values from your level.




Sample 2 (image 5): lesson card shown after the storm. The lesson line can be reworded to match your level.

## 2. Video clips (3-5 seconds each, muted under your voice)

| # | Clip idea | Use in the video | Where to get it |
|---|---|---|---|
| **1** | NASA animation of the 2018 dust storm spreading across Mars | Opening clip while you say the storm hits. | PIA22519 (storm growing) or PIA22487 (before/after)<br>Credit: NASA/JPL-Caltech/MSSS |
| **2** | NASA map of dust blocking light (tau) over time, with Opportunity marked | Adds the real NASA data feeling. Shows how much light the dust blocks. | PIA22737<br>Credit: NASA/JPL-Caltech |
| **3** | Screen recording or animation of the game: lights switching off one by one while the battery bar drops | The core gameplay moment. Shows the player making the choice. | Record from your prototype, or animate Sample 1 in Canva or PowerPoint. Made by the team. |
| **4** | Hand flipping a switch, or warning lights flickering on a control panel | Cutaway for “what do you switch off?” | Free stock on Pexels (search: flickering lights, control panel warning). Credit it. |
| **5** | Mars surface or dust-devil footage from a NASA rover | Calm-before-the-storm shot, or a bridge between clips. | NASA Mars Exploration Program media. Check NASA’s media usage rules and credit it. |

Sourcing note: use NASA’s own media (with the credit lines above) or free sources like Pexels. The challenge requires crediting all third-party assets and AI, so add every source to the team credit list as you collect it. Samples 1 and 2 are drawn by the team, not taken from NASA.

## 3. Key points and roadmap options (pick one)

All three options use the same 40 seconds and end with the same lesson line, so the choice is about how the story is told.

| Option | Key points (in order) | Why choose it | Watch out for |
|---|---|---|---|
| **A. The Countdown** | Sun fades, battery drops, "what do you switch off?", result, lesson. | Feels like a real game moment. Easy for kids and judges to follow. Builds tension fast. | Needs a clear, good-looking decision screen. |
| **B. Real Storm, Real Data** | NASA storm footage, Opportunity's faint Sun, "your outpost faces the same storm", game choice, lesson. | Strongest link to NASA data. Real images are easy to source. | Can feel like a documentary. Keep at least 15 seconds on the game. |
| **C. Two Choices, Two Endings** | Same storm, split screen: switch off comms vs switch off the greenhouse, two different results, lesson. | Shows the trade-off idea directly. Matches the tagline "Every mission is a trade-off." | Needs two quick mock screens. 40 seconds is tight. |

**Suggested pick:** Option A with a NASA opening from Option B. Real NASA footage in the first 16 seconds gives credibility, then the game takes over for the choice and the lesson. Option C can become a short extra line if time allows.

| Time | Visual | Say (draft) |
|---|---|---|
| **2:40-2:48** | Clip 1 + Image 1 (storm spreading) | "Then the storm hits. Dust spreads across the planet and hides the sun." |
| **2:48-2:56** | Image 2 + Clip 3 (battery dropping) | "The solar panels go dark. The batteries start to drop." |
| **2:56-3:06** | Image 4 + Clip 4 (decision screen) | "You are the Commander. Lights, heating, comms, rover charging. What do you switch off?" |
| **3:06-3:14** | Clip 3 (result) | "Every choice saves power, and costs something else." |
| **3:14-3:20** | Image 5 (lesson card) | "Then the game shows what went wrong, and why." |

***Timing tip:*** *the draft is about 75 words, so read it aloud with a timer. Under 36 seconds, slow down and pause after the decision question. Over 40 seconds, cut the line at 3:06.*

## 4. Image and animation sequence

Add the assets in this order, following the draft script above. Times are targets; adjust them to your voice recording, but keep the end at 3:20 or earlier.

| Time | Asset and format | Animation and transition | Source |
|---|---|---|---|
| **2:40-2:48** | Clip 1, MP4, 1080p, 30 fps, 3-4 s, muted.<br>Then Image 1 (before/after), PNG or JPG, 1920x1080. | 0.5 s cross-dissolve in from the Part 4 certificate.<br>Slow zoom into the planet during the clip.<br>Image 1: slow wipe from the May view to the July view.<br>Credit tag "NASA/JPL-Caltech/MSSS" in the lower corner for 2 s. | NASA (PIA22519 or PIA22487) |

| Time | Asset and format | Animation and transition | Source |
|---|---|---|---|
| **2:48-2:56** | Image 2 (Opportunity's last images), then Clip 3 (prototype or Sample 1 animation), MP4. | Image 2 fades toward black as you say "go dark".<br>Dissolve to Clip 3: the battery bar falls and turns orange; red circle around the power chart dropping.<br>Lower-third tag: "Current prototype: 30-sol mission" if this is real footage.<br>Optional 2 s cutaway to Image 3 (dusty panel). | Image 2: NASA (PIA22930, PIA22929).<br>Clip 3: your own recording |
| **2:56-3:06** | Image 4 (decision screen, Sample 1), PNG.<br>Clip 4 (switch or warning lights), MP4, 2 s, muted. | Decision screen slides up from the bottom.<br>The question "What do you switch off?" pulses once.<br>Switches flip off one at a time (0.5 s each) as you say each item.<br>Cut to Clip 4 for the last 2 s. | Image 4: you make it.<br>Clip 4: Pexels or Unsplash |
| **3:06-3:14** | Clip 3 (result), MP4. | The battery line drops more slowly after the switches go off.<br>The "saves 18% \| slower food growth" style chips highlight one at a time, matching "saves power, and costs something else".<br>Optional: 2 s of the real controls panel (heater slider, greenhouse lights, Why? button). | Your own recording or animation |
| **3:14-3:20** | Image 5 (lesson card, Sample 2), PNG. | The two battery lines draw in (red first, then green).<br>Lesson badges 1, 2, 3 pop in (0.3 s each).<br>Hold still for 1.5 s, then dissolve into Part 6. | You make it |

***Unused items:*** *Image 3, Clip 2 (tau map) and Clip 5 (rover footage) are optional extras. Use them only if the timing allows.*

## Note on wording

The Sample 1 decision screen lists "Earth communications" and "Rover charging" as switches. The current prototype does not have those; its live controls are the heater set point, greenhouse lights, ice miner, panel cleaning, shelter and repair. Either label the screen "Concept mock-up", or swap the switches for real controls (greenhouse lights, heater, ice miner) and change the script line to match. The numbers in Sample 1 are placeholders, so do not present them as real. The NASA storm images are real, while the storm effect in the game is a model, so show a DERIVED label if you show game values.

# Part 6: Impact and Needs (3:20-4:00)

about 30 seconds of narration, plus the credits. Goal: what this will change, what you need next, and a close that answers the opening question. This part must end by 3:58 because the 4:00 limit is a hard cutoff.

## 1. Images (one every 7-12 seconds)

- A before/after graphic: a grey "Game over" screen with no explanation on the left, the colorful lesson card from Part 5 (Sample 2) on the right. Make the grey screen yourself and reuse the lesson card.
- A two-icon "what we need" slide: a data-layers icon labeled "Official NASA data" and a group-of-students icon labeled "Playtesters, ages 10-14". Make it yourself.
- Closing card: the Part 1 question "How do astronauts survive on another planet?" over the Part 2 split image (dream | NASA data). Reuse your own files.
- Credits card with team names and sources. Make it yourself.

## 2. Video clips (3-4 seconds each, muted under your voice)

- A kid on a tablet reacting with excitement (free stock). Used in the impact line.
- Students working together on laptops in a classroom (free stock). Used with the playtesting need.

## 3. Script (about 30 seconds)

| # | Image cue | Say |
|---|---|---|
| **1** | Before/after graphic + tablet clip | "A child does not just hear that Mars is hard. They see why, and learn that a failed design is information, not the end." |
| **2** | Two-icon "what we need" slide + classroom clip | "We need the official NASA data to replace our placeholder numbers, and students aged ten to fourteen to playtest it. We haven't tested with students yet." |
| **3** | Closing card, then credits | "How do astronauts survive on another planet? By making smart trade-offs. Now kids can practice them. Thank you." |

***Timing tip:*** *read it aloud with a timer. The script is about 67 words and should take 30 to 33 seconds. Under 28 seconds, slow down and pause after line 1. Over 35 seconds, cut "Now kids can practice them." from line 3. Do not cut the "not tested" line.*

## 4. Image and animation sequence

Add the assets in this order. Times are targets; adjust them to your recording, but keep the end at 3:58 or earlier.

| Time | Asset and format | Animation and transition | Source |
|---|---|---|---|
| **3:20-3:31** | Static PNG, 1920x1080.<br>Before/after graphic.<br>Plus tablet clip, MP4, 3-4 s, muted. | 0.5 s dissolve in from the Part 5 lesson card.<br>Left side (grey "Game over") gets a red X, then the right side (lesson card) slides in.<br>Cross-dissolve to the tablet clip for the last 4 s. | Graphic: you make it, reusing the Part 5 lesson card.<br>Clip: Pexels or Unsplash |
| **3:31-3:44** | Static PNG, 1920x1080.<br>Two-icon "what we need" slide.<br>Plus classroom clip, MP4, 3-4 s, muted. | Data icon pops in on "official NASA data"; student icon pops in on "students aged ten to fourteen".<br>Chip appears on "haven't tested": "Not yet tested with students".<br>Cross-dissolve to the classroom clip for the last 3 s. | Slide: you make it.<br>Clip: Pexels or Unsplash |
| **3:44-3:50** | Static PNG, 1920x1080.<br>Closing card: Part 1 question over the Part 2 split image. | The question types on again, as in Part 1, while you say it.<br>At "smart trade-offs" the answer fades in below: "Smart trade-offs."<br>Slow zoom-out (108% to 100%). | Your own files from Parts 1 and 2 |
| **3:50-3:58** | Static PNG, 1920x1080.<br>Credits card. | Fade in while you say the last words, then hold still for at least 4 s so it is readable. No scrolling. Fade to black at 3:58. | You make it |

***Format rules:*** *export everything at 1920x1080 (16:9). Use one font and the same navy and orange palette as the other parts. Keep every transition under 0.5 s. Mute all clip audio under your voice.*

## Bonus: Pinterest searches (for ideas only)

**Photos:** game over screen vs lesson screen UI, students testing a learning game, credits card design, space mission title card.

**Videos:** kid playing tablet game, students working on laptops classroom.

## Sourcing note

The grey game-over screen, the "what we need" slide and the credits card are your own work, so list the design tool you used. Take stock clips from Pexels or Unsplash. Do not put the NASA logo on any slide; write "NASA data" as plain text instead.

## Note on wording

The learning outcomes are design goals, not measured results, which is why line 2 says the game has not been tested with students. Parts 3 and 4 describe planned features, so do not imply them in this part.

# Master credits list (all parts)

The on-screen credits card has only 8 seconds, so show the short form on screen and put the full list in the submission description. Check the official rules for the exact credit format.

## Short form for the on-screen card (3:50-3:58)

- Team: [all team member names]. Every member must be named.
- NASA Curiosity RAD data, NASA/USGS SWIM, NASA Mars Trek
- NASA/JPL-Caltech imagery (MSSS, Cornell, ASU); three.js (cdnjs)
- Stock clips: Pexels and Unsplash
- [AI tools used, e.g. “Script drafting assisted by Claude (Anthropic)”]

## Full list for the submission description

| Part | Asset | Source and credit |
|---|---|---|
| 1 | Stars or Milky Way time-lapse | [creator, Pexels or Unsplash link] |
| 1 | Astronaut spacewalk or station footage | NASA media library [title and link] |
| 1, 6 | Question card, word card, seesaw, wordmark, team card, grey game-over screen, what-we-need slide, credits card | Made by the team in [tool] |
| 2 | Child, helmet, classroom, rocket launch and tablet images and clips | [creator and link for each] |
| 2 | Curiosity view of Gale Crater, Mars dust storm image, rover footage | NASA [title and link for each] |
| 2, 6 | Split image: dream \| NASA data chart | Made by the team; NASA chart [credit] |
| 3 | Title card, crew illustration, dashboard sketch, staircase | Made by the team; crew art note the AI tool and what it made |
| 3 | Red warning light clip; dim, dusty Mars footage | Stock [creator and link]; NASA [title and link] |
| 4 | Gale Crater photo, rover footage | NASA [title and link] |
| 4 | Site cards, budget screen, level icons, crew portraits, event icons, patch, badge, certificate | Made by the team in [tool]; note any AI use |
| 5 | PIA22487, PIA22519 (2018 dust storm) | NASA/JPL-Caltech/MSSS |
| 5 | PIA22930, PIA22929 (Opportunity’s last images) | NASA/JPL-Caltech/Cornell/ASU |
| 5 | PIA22737 (dust opacity map), dusty panel image, rover footage | NASA/JPL-Caltech; [others, title and link] |
| 5 | Decision screen (Sample 1), lesson card (Sample 2) | Made by the team |

| Part | Asset | Source and credit |
|---|---|---|
| 5, 6 | Switch or warning light clip; tablet and classroom clips | Pexels [creator and link] |
| All | Game footage | Made by the team; three.js r128 (cdnjs) |
| All | Music, fonts, AI tools | [add each with source] |